

Large meta-analysis of genome-wide association studies expands knowledge of the genetic etiology of Alzheimer's disease and highlights potential translational opportunities

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ABSTRACT

Deciphering the genetic landscape of Alzheimer's disease (AD) is essential to define the pathophysiological pathways involved and to successfully translate genomics to potential tailored medical care. To generate the most complete knowledge of the AD genetics, we developed through the European Alzheimer's Disease BioBank (EADB) consortium a discovery meta-analysis of genome-wide association studies (GWAS) based on a new large case-control study and previous GWAS (in total 39,106 clinically diagnosed cases, 46,828 proxy-AD cases and 401,577 controls) with the most promising signals followed-up in independent samples (18,063 cases and 23,207 controls). In addition to 34 known AD loci, we report here the genome-wide significant association of 31 new loci with the risk of AD. Pathway-enrichment analyses strongly indicated the involvement of gene sets related to amyloid and Tau, but also highlighted microglia, in which increased gene expression corresponds to more significant AD risk. In addition, we successfully prioritized candidate genes in the majority of our new loci, with nine being primarily expressed in microglia. Finally, we observed that a polygenic risk score generated from this new genetic landscape was strongly associated with the risk of progression from mild cognitive impairment (MCI) to dementia (4,609 MCI cases of whom 1,532 converted to dementia), independently of age and the *APOE* ϵ 4 allele.

INTRODUCTION

The objectives of translational genomics and subsequent personalized medicine are to define prognosis/diagnosis markers of a disease and to adapt treatments at the individual level. Such approaches have already been successful in cancers and promising approaches are emerging for tailored treatments for diabetes¹. However, in many other pathologies such as dementia, personalized approaches are still a concept and have proven difficult to develop.

Indeed, dementia is a term that encompasses an array of complex phenotypes in which the main symptom is the progressive decline of cognitive performance as observed in Alzheimer's disease (AD), the most common type. The dementia stage in AD is the culmination of a long, progressive, and silent process that is followed by intermediate pathological changes leading ultimately to cognitive decline and dementia. Importantly, the concept of AD has changed to recognize that AD is a continuum with a long preclinical phase of Subjective Cognitive Decline (SCD), a stage of mild cognitive impairment (MCI), and a dementia phase. The preclinical phase may offer unique opportunities for prevention of AD through early detection of AD pathology and application of pharmacological treatment with disease-modifying drugs that are still under development.

In this context, translational genomics may be of particular interest in AD since this disease exhibits a particularly high heritability, estimated between 60 and 80%². Indeed, since the first large genome-wide association studies (GWAS) published in 2009^{3,4}, many loci/genes have been associated with the risk of developing AD⁵. These genetic findings have contributed to the identification of pathways and networks underlying AD, specifically implicating immunity, cholesterol processing, endocytosis, and more recently, the role of A β and Tau in the pathogenesis of common forms of AD⁶. Indeed, it is expected that the discovery of genetic risk factors for AD will reveal additional relevant pathogenic pathways operating in AD. Mounting evidence now suggests that AD is a disease in which multiple components combine to trigger the disease, beyond the dominant "amyloid cascade hypothesis". This observation may also indicate that preponderant deleterious pathway(s) might be differentially involved at the individual level. For instance, therapies targeting the APP metabolism pathway may not be effective if this pathway is relatively unimportant for an individual with a particular genetic profile. If true, this would imply the importance of pursuing multiple therapies that target different genetically driven-pathways. Once a range of treatment options become available, an individualized model of AD pathology would be feasible, such that polytherapies and personalised medicine approaches can be developed and applied.

In a personalized medicine framework, it will be essential to translate large-scale genomic information into useful tools for personalized risk prediction and subsequent potential tailored intervention, for example through polygenic risk scores (PRS). Generating PRSs has been regarded as a reasonable solution to summarize genome-wide genotype data into a single variable that measures the genetic contribution to a trait or a disease for a particular individual. Herein, genomic information offers a unique opportunity for early detection. However, a large part of the AD genetic component is still unknown and several loci/genes also need to be confirmed as genuine genetic risk factors.

As a consequence, strong efforts are still needed to characterize the genetic architecture of AD, with the objectives to identify critical pathways and construct powerful PRSs for the disease. Within this background, increasing the size of GWAS data is an obvious way to facilitate the characterization of new genetic risk factors as observed in many other multifactorial diseases. In addition, since rare variants might explain a large proportion of the missing heritability, improving their analyses is also mandatory. Taking into account these two major points, we developed the European Alzheimer's Disease BioBank (EADB) consortium grouping together the main European GWAS consortia already working on AD and a new dataset of 20,464 AD cases and 22,244 controls collated from 15 European

countries (Belgium, Bulgaria, Czech Republic, Denmark, Finland, France, Germany, Greece, Italy, Portugal, Spain, Sweden, Switzerland, The Netherlands and The United Kingdom). We then benefited from the Trans-Omics for Precision Medicine (TOPMed) imputation panel based on whole-genome sequencing of 62,784 individuals in order to increase the number of variants tested and to improve the imputation quality of the rare variants⁷. The EADB results were meta-analysed with a proxy-AD GWAS performed in the UK Biobank. The best hits ($p \leq 10^{-5}$) generated from this meta-analysis step were then replicated in a large set of samples from the ADGC and CHARGE consortia.

In addition, in EADB, we collected an independent longitudinal cohort of 4,609 MCI cases of whom 1,532 converted to dementia. This provided us with the unique opportunity to test the association between a PRS we generated from our GWAS data and the risk of progression to dementia/AD with the objective to translate genomic information into personalized risk profiles for early detection of AD risk.

RESULTS

GWAS analysis

The EADB Stage I (GWAS meta-analysis) was based on 39,106 clinically diagnosed cases, 46,828 proxy-AD cases and 401,577 controls (Supplementary Tables 1 and 2) and on 21,101,114 variants after quality control. Genomic inflation factors (λ) were slightly inflated (λ = 1.08 overall and 1.17 when restricted to variants with minor allele frequency (MAF) above 1% (see Supplementary Figure 1 for a quantile–quantile (QQ) plot). However, linkage disequilibrium score (LDSC) regression estimate⁸ indicated that the majority of this inflation was due to a polygenic signal, with the intercept being close to 1 (intercept=1.05, s.e=0.01 versus λ =1.2 on the variants considered in the LDSC analysis).

We selected all variants with P value less than 1×10^{-5} in Stage I. We defined non-overlapping regions around those variants, excluded the region corresponding to *APOE*, and sent the remaining variants for follow-up (Stage II, n =11,390, see Methods) in a large set of samples from the ADGC and CHARGE consortia (18,063 cases and 23,207 controls). A signal was considered as genome-wide significant if nominally replicated (P values ≤ 0.05) in the same direction in the Stage I and Stage II analyses and if associated with AD risk with a P value less than 5×10^{-8} in the Stage I + Stage II meta-analysis. In addition, we applied a PLINK clumping procedure⁹ to define potential independent hits from the Stage I results (see Methods). After validation by conditional analyses (see Supplementary Information and Supplementary Tables 3 and 4), this approach led us to confirm 38 signals in 34 loci already known to be associated with the risk of developing AD in the main previous AD GWASs^{6,10–15} (Table 1) and to propose 31 new loci (Table 2 and Supplementary Figures 2–24). Five of these loci (*APP*, *CCDC6*, *NCK2*, *TSPAN14* and *Shapin*) were already reported in two preprints using GWAS data included in our study^{16,17}. Besides, the *NDUFAF6* and *IGH* loci were previously reported in a gene-wide analysis¹⁸. Of note, the magnitudes of associations in Stage I were highly similar to those observed if we restricted the Stage I to clinically-diagnosed cases, hereafter denoted as diagnosed cases only analysis (Supplementary Table 5 and Supplementary Figure 25). In addition, we did not detect any signal which may be mainly brought by the proxy-AD cases (Supplementary Table 5). Of note, we also provided a list of four loci with a genome-wide significant signal in the Stage I + Stage II analysis, but failing in Stage II (Supplementary Table 6).

Pathway analyses

To evaluate the biological significance of this new AD genetic landscape, we first performed pathway enrichment analyses. 150 gene sets were significant after multiple testing correction ($q \leq 0.05$, see Methods) in the Stage I (Supplementary Table 7), with the 20 most significant pathways shown in Table 3. The most significant gene sets relate to amyloid and Tau, with many of the other significant gene sets relating to lipids and immunity. Notably, there are gene sets related to macrophage activation and microglial cell activation. We also assessed whether enrichment pathway analyses were sensitive to the inclusion of the proxy-AD cases in Stage I. When analyses were performed limited to the diagnosed cases only, 69 gene sets were significant ($q \leq 0.05$) (Supplementary Table 7). Of these 69 gene sets, 53 reached $q \leq 0.05$ and all 69 reached $p \leq 0.05$ in the full Stage I that includes proxy-AD cases. This indicates that including proxy-AD cases did not mask disease-relevant biological information. We also repeated the enrichment analysis by using a window of 35kb upstream and 10kb downstream to assign variants to genes or by removing all 70 genes within 1Mb of *APOE*. Results were consistent between analyses (Supplementary Table 7).

We then performed a single-cell expression enrichment analysis in MAGMA using human data from the Allen Brain Atlas dataset (see Methods). Two complementary measures were used for each cell type: average gene expression per nucleus (Av. exp) and percentage of nuclei in a cell type expressing each gene (% Cell exp). Gabaergic/glutamergic neurons, astrocytes, oligodendrocytes, microglia and endothelial cells were analyzed and only

microglia expression reached significance after correcting for multiple testing in the two measurements (FDR $\leq$ 0.05; Av. Exp.; $p=4.60\times 10^{-4}$ and % Cell. Exp. $p=5.59\times 10^{-8}$, Supplementary Table 8), with increased expression corresponding to more significant association with AD risk. A similar result was also observed using the mouse single cell dataset from Skene et al¹⁹ (Supplementary Table 9).

We tested whether the relationship between microglia expression and association with AD risk was specific to particular areas of biology by using MAGMA. In particular, we tested for interaction between expression and pathway membership of each of the 91 significant pathways containing at least 25 genes with measured expression (Supplementary Table 10). Among several significant interactions, the most significant one was detected between GO:1902991 (regulation of amyloid precursor protein catabolic process) and gene expression level in microglia after multiple testing correction (q-values= 1.2×10^{-12} and 8.3×10^{-8} for % Cell Exp. and Av. Exp. respectively). This interaction is still significant when the APOE locus is removed (q-values= 3.8×10^{-11} and 7.8×10^{-4} for % Cell Exp and Av. Exp. respectively). This observation indicates that among the amyloid-beta gene sets showing an overall enrichment for AD association signal, it is the genes in these pathways with highest microglia expression that show the most association, suggesting a functional relationship between microglia and APP/A β pathways. A complete list of genes in APP/A β pathways GO:1902991 (but also GO:1902003), along with their GWAS significance and microglia expression, is given in Supplementary Table 11.

Gene prioritization

In order to prioritize candidate genes in the new loci, we considered the “nearest” gene from the lead variant and the genes exhibiting AD-related modulations within a region of 1 Mb around the lead variant according to those criteria: (i) expression and splicing quantitative trait loci (eQTLs and sQTLs) and colocalization analyses combined with transcriptome-wide association studies on expression and splicing (eTWAS and sTWAS) in AD-relevant brain regions; (ii) genetic-driven methylation as a biological mediator of genetic signals in blood (MetaMeth). We also considered those additional criteria: (i) the functional impact of gene under-expression on APP metabolism²⁰; (ii) methylation QTL (mQTL) and histone acetylation QTL (haQTL) effects of the lead variants in dorsolateral prefrontal cortex (DLPFC)²¹ and (iii) additional eQTL effects of these variants in naïve state monocytes and macrophages^{22–27}. All the results were summarized in Fig. 2 and a full description of how the genes were prioritized is reported in the Supplementary Information (see also Supplementary Tables 12-21 and Supplementary Fig. 26-34).

Although the lead variants did not fall within a gene, for eight of the novel loci, brain molecular QTL, TWAS, blood MetaMeth and/or APP metabolism results exclusively supported the genes nearest to the lead variant: *OTULIN* (locus 4), *RASA1* (locus 5), *ICA1* (locus 9), *TMEM106B* (locus 10), *ABCA1* (locus 15), *CTSH* (locus 23), *MAF* (locus 25) and *SIGLEC11* (locus 28). Three other “nearest” genes in the new loci can be prioritized since the lead variant corresponds to a predicted deleterious missense variant within the gene itself: *MME* (locus 2), *FDFT1* (locus 12), and *SHARPIN* (locus 14). For *SHARPIN*, we found additional evidence that AD risk in the locus is associated with SHARPIN expression and splicing events (Supplementary Tables 17-18 and Supplementary Fig. 31-32). Finally, for six loci: *NCK2* (locus 1), *RASGEF1C* (locus 6), *HS3HT5* (locus 7), *UMAD1* (locus 8), *C1S* (locus 19) and *APP* (locus 31), none of the candidate genes could be prioritized based on genetic-driven expression, splicing or methylation analyses, therefore we considered that their proximity to the lead variant was in favor of their prioritization but at this stage at a lower level of confidence. Of note, *APP* is an obvious candidate gene, but *CYYR1-AS1* in between *APP* and *ADAMTS1* might also be of interest (Supplementary Tables 17, 21 and 31).

The remaining 14 novel loci present a more complicated pattern; several genes exhibit AD-related modulations in the same locus, and/or the prioritized gene is not the nearest protein coding gene. First, we could efficiently prioritize candidate risk genes in 4 additional loci:

EGFR (locus 11), *TSPAN14* (locus 17), *BLNK* (locus 18) and *GRN* (locus 27). For instance, in locus 11, *EGFR* is a likely candidate gene because its eQTL signals colocalize with the AD risk association signal, and its fine-mapped eTWAS hits (with FOCUS PIP values of ~1) associate predicted increased *EGFR* expression with increased AD risk (Fig 3; Supplementary Tables 15,17; supplementary Fig. 29, 31). In the complex locus 17, *TSPAN14* was identified as the candidate risk gene as it exhibited numerous AD-related expression and splicing modulations, including novel cryptic complex splicing events that we identified and experimentally confirmed (Fig. 3; Supplementary Tables 13-19 and supplementary Fig. 26,27,29-32).

We did not clearly identify a single candidate in the remaining 10 loci. However, current evidence points towards the following candidate genes: i) *DGKQ*, *SCL26A1* and *IDUA* in the complex locus 3; ii) *CCDC6* in locus 16; iii) *SNX1* in locus 22 (importantly, we previously determined that *SNX1* and *APH1B* GWAS signals in this region were independent, see Supplementary Tables 3 and 4); iv) *INO80E*, *DOC2A*, *TBX6* and *YPEL3* in the most complex new locus 24 (*ALDOA* locus); v) *LIME1* and *RTEL* in locus 30. In locus 13, among the 4 genes, *NDUFAF6* exhibited the highest number of hits but *TP53INP1* was also of interest. In locus 21, a recent report pointing to an increased burden of rare variants in *ATP8B4* in AD patients prioritizes this gene²⁸. For the locus 29, we consider *LILRB2* as a plausible risk gene according to bibliographical data^{29,30}. Finally, we were not able to prioritize a gene in the complex IGH cluster (locus 20), nor in the *PRDM7* locus (locus 26).

Polygenic risk score

In order to explore the effect of a genetic burden on progression from mild cognitive impairment to AD-dementia, a PRS based on the genetic data generated above (see Methods and Supplementary Table 22) was constructed and tested in several longitudinal cohorts of MCI cases (Supplementary Table 23).

We observed a significant association of this PRS with the risk of progressing to any type of dementia (HR=1.028 per average risk variant, 95%CI [1.022-1.033], $p=4.93 \times 10^{-7}$) or with the risk of progression to AD dementia only (HR=1.033 per average risk variants, 95%CI (1.027-1.039), $p=8.6 \times 10^{-8}$) after adjustment on age, sex, principal components and the number of APOE-e4 alleles (Figure 4). Unadjusted analysis, analysis adjusted for age, sex and PCs only and coding non-AD converters as censored cases in the progression to AD dementia analysis did not change the results (see Supplementary table 24). Importantly, association of the PRS with progression risk does not seem to be modified by the presence of APOE-ε4 since we did not find any significant interaction between the number of APOE-e4 alleles and the PRS whatever the model tested.

Of note, the number of APOE-e4 alleles itself exerted a strong effect on the progression to all-cause of dementia (HR=1.64, 95%CI [1.51-1.78], $p=1.2 \times 10^{-33}$) and AD-dementia (HR=1.79 [1.64-1.96], $p=3.6 \times 10^{-38}$). This effect corresponds to carry 18 average risk variants coded in our PRS.

DISCUSSION

This meta-analysis combining a new large case-control study and previous GWAS identified 69 independent genetic risk factors for AD, 38 previously reported in published GWAS and 31 corresponding to novel signals, strongly expanding our knowledge of the complex genetic landscape of Alzheimer's disease. We leveraged genetic, functional, and prior literature to nominate credible candidate genes at each new locus with a particular emphasis on transcriptional regulation, methylation, and APP metabolism datasets. A short description of these genes and their potential implication in AD are described in the Supplementary Information.

Remarkably, our meta-analysis and the characterization of these new loci clarify the global picture of AD etiology. For instance, pathway enrichment analyses remove ambiguities concerning the involvement of Tau binding proteins and APP/A β metabolism as major actors of the AD processes, beyond the levels of certainty previously described⁶.

In addition, beyond the genetic risk factors already known to be involved in the APP metabolism, i.e. *ADAM10*, *APH1B* and *FERMT2*²⁰, we also proposed five candidate genes susceptible to modulate this metabolism (*DGKQ*, *RASA1*, *ICA1*, *DOC2A* and *LIME1*). Although further investigations are needed to determine their exact implications, our data clearly support the central role of the APP/A β metabolism/functions in the pathophysiological process of late-onset forms of AD. Of note, none of these genes were included in the GO:1902992 pathway (negative regulation of amyloid precursor protein catabolic process) we characterized as the most enriched in potential AD genetic risk factors (Table 3).

These enrichment pathway analyses also confirmed the involvement of innate immunity and microglial activation in AD (Table 3). In addition, single-cell expression enrichment analysis also highlighted genes expressed in microglia (Supplementary Tables 8 and 9). Finally, 9 of our prioritized genes, i.e. *OTULIN*, *RASGEF1C*, *TSPAN14*, *BLNK*, *ATP8B4*, *MAF*, *GRN*, *SIGLE11C* and *LILRB2*, appeared to be mainly or almost only expressed in microglia (Fig. 2). However, at this time, only *GRN* is currently referenced in the microglial cell activation pathway (GO:0001774; Table 3). This suggests that enrichment of pathways involving microglia may be underestimated in our current analysis and further works will be clearly needed to determine whether and how these 9 genes may be involved in microglia function/activation. Several publications have already demonstrated involvement of the corresponding proteins in microglia function/activation (see Supplementary Information) and importantly three, i.e. *GRN*, *SIGLEC11* and *LILRB2* have also been linked to A β peptides/amyloid plaques^{29,31,32}. Taking into account the already known genetic risk factors primarily expressed in microglia, i.e. *INPPD5*, *TREM2*, *SPI1*, *MS4A4A*, *SPPL2A*, *PLC γ 2* and *ABI3* (Supplementary Figure 35), this means that at least 25% of the genetic risk loci described in this paper are credibly linked to AD-related microglia dysfunctions. Importantly, *TREM2*, *PLC γ 2*, *ABI3* and *INPP5D* were also characterized as microglia A β response proteins at the transcript and/or protein levels³³. This observation thus indicates that at least 7 genes (44% of those mostly expressed in microglia) have been already linked to A β clearance/toxicity. However, it is also necessary to keep in mind that gene prioritization, while efficient for numerous loci, presents some limitations, particularly in complex locus where it is difficult to clearly identify the most relevant gene. In addition, it is important to note that for our molecular QTL-based analyses, we only considered cis-QTLs that are typically found within 1 Mb around the molecular phenotype feature. Therefore our analyses might have missed certain molecular trans-QTLs with important effect on AD risk. Furthermore, even though we extensively integrated our GWAS results with the information derived from expression, splicing, and methylation landscape of newly identified genetic risk regions that hinted at possible explanation for mechanism of action of the AD association in these regions, our post-GWAS analyses did not account for the possibility that the underlying risk mechanism could be explained through the effect of genetic variation on protein levels (protein QTLs), metabolite levels (metabolite QTLs), 3D spatial organization of chromatin (e.g. topologically associated domains [TADs]) or a structural variant that might be tagged by

the newly identified genetic risk variants. We therefore emphasize that for complete elucidation of AD risk mechanisms in these regions, more investigations are required. We therefore cannot exclude that we overstated some pathophysiological pathways based on incorrect gene assignment or incomplete information. Nevertheless, reinforcing the role of microglia in AD, our data also revealed for the first time a (direct or indirect) statistical relationship between gene expression in microglia, genetic risk factors and APP/A β pathways (Supplementary table 10).

Translating genetic findings into tools that can be used in the clinical setting has proven to be challenging because different strategies are available to create a PRS summarizing personal genetic burden. In our study, we computed a PRS following the strategy previously described by Chouraki et al³⁴ in which increase of one point on the PRS corresponds to carrying one additional average risk allele. We observed that this PRS was associated with the risk of progression from MCI to dementia and to AD dementia (ADD), the major form of dementia. Importantly, the association was obtained while the *APOE- ϵ 4* allele was not included in the PRS. Moreover, the association did not disappear following adjustment for age or additive effect of the *APOE- ϵ 4* allele. Previous studies evaluating the effect of PRS on progression of MCI to ADD have provided compelling evidence supporting the role of *APOE*, whereas the contribution of additional genetic variants to progression has not shown unanimous results³⁵. Our finding of an *APOE*-independent effect on MCI progression may be explained by (i) the large longitudinal sample of 4,609 MCI cases of whom 1,532 converted to dementia (ii) and the improved knowledge of the genetic component of AD through the larger number of newly-discovered genome-wide significant variants included in the PRS. Our study also shows that carrying 18 "average risk variants" conferred a similar probability of progressing to ADD as being *APOE- ϵ 4* heterozygous, while carrying 32 "average risk variants" resemble that of being *APOE- ϵ 4* homozygous. Although further research is still needed before these findings can be translated into the clinical routine, completing the genetic architecture of AD is definitely paving the way to personalized risk prediction even before dementia stage is reached.

Of note, several new AD loci were also associated with the risk of developing other neurodegenerative diseases: *IDUA* locus with Parkinson's Disease (PD), *GRN* and *TMEM106B* loci with fronto-temporal dementia (FTD). According to the large number of cases analyzed in our study and the well documented clinical diagnostic errors between neurodegenerative diseases, we cannot exclude that these associations are in part due to contaminations by PD or FTD cases in our sample. However, GWAS colocalization analyses indicate that the main signal in PD is independent of the one observed in AD (Supplementary Tables 25 and 26). Only small GWAS are available in FTD (and depending on the type of FTD)³⁶, this makes difficult to definitely answer to the independency of the AD and FTD signals. Further investigations will be required all the more since sporadic FTD has been described as a polygenic disorder where multiple pleiotropic loci with small effects contribute to increased disease risk³⁷. In addition, the lead *GRN* variant in AD is functional and has also been described to be associated with TDP-43 positive FTD risk³⁸. In this context, it will be interesting to determine whether our PRS may be specific or not of AD given the potential genetic overlap between neurodegenerative diseases. Of note, our PRS was not associated with risk of converting to non-AD dementia (HR=1.010, 95%CI [0.996-1.025], $p=4.8 \times 10^{-1}$). However, this absence of association may be due to lack of statistical power in our current analysis and will require further investigations.

In conclusion, our work demonstrates that improved characterization AD genetics also expands our knowledge of the underlying pathophysiological processes, presenting novel opportunities for therapeutic approaches and risk prediction through robust PRS. Convergence between treatments generated from genomics and PRS may thus pave the way to translational genomics and personalized medicine.

MATERIALS AND METHODS

Samples. All discovery meta-analysis samples are from the following consortia/datasets: EADB, GR@ACE, EADI, GERAD/PERADES, DemGene, Bonn, the Rotterdam study, the CCHS study, NxC and the UK Biobank. Summary demographics of these case-control studies are described in Supplementary Table 1 and more detailed descriptions are available in the Supplementary Information. Replication samples are from the ADGC and CHARGE consortia (Supplementary Table 1 and Supplementary Information) and fully described elsewhere^{6,11,14,15,39–41}. Written informed consent was obtained from study participants or, for those with substantial cognitive impairment, from a caregiver, legal guardian, or other proxy. Study protocols for all cohorts were reviewed and approved by the appropriate institutional review boards. Further details of all cohorts can be found in the Supplementary Information.

Quality control and imputation. Standard quality control was performed on variants and samples on all datasets individually. The samples were then imputed with the Trans-Omics for Precision Medicine (TOPMed) reference panel^{42,43}. The Haplotype Reference Consortium (HRC) panel⁴⁴ was also used for some datasets (Supplementary Table 2). For the UK Biobank, we used the provided imputed data, generated from a combination of the 1000 Genomes (1000G), HRC and UK10K reference panels. See Supplementary Information for more details.

Stage I analyses. Association tests between AD clinical or proxy status and autosomal genetic variant were conducted separately in each dataset using logistic regression assuming an additive genetic model as implemented in SNPTEST⁴⁵ or in PLINK⁹, except in the UK Biobank where a logistic mixed model as implemented in SAIGE⁴⁶ was considered. Analyses were performed on the genotype probabilities in SNPTEST (newml method) and on dosage in PLINK and SAIGE. Analyses were adjusted for principal components and genotyping centers when necessary (Supplementary Table 2). For the UK Biobank dataset, effect sizes and standard errors were corrected by a factor of two to take into account that proxy cases were analysed¹². We filtered out duplicated variants and variants with (i) missing effect size, standard error or P value, (ii) absolute value of effect size above 5, (iii) imputation quality less than 0.3, (iv) the product of the minor allele count and the imputation quality (mac-info score) less than 20. In the UK Biobank dataset, only variants with minor allele frequency (MAF) above 0.01% were analyzed. For datasets not imputed with the TOPMed reference panel, we also excluded (i) variants for which conversion of position or alleles from the GRCh37 assembly to the GRCh38 assembly was not possible or problematic, or (ii) variants with very large difference of frequency between the TOPMed reference panel and the reference panels used to perform imputation.

Results were then combined across studies with a fixed-effect meta-analysis using the inverse variance weighted approach as implemented in the METAL software⁴⁷. We filtered (i) variants with heterogeneity P value below 5×10^{-8} , (ii) variants analyzed in less than 20% of the total number of cases and (iii) variants with frequency amplitude above 0.4 (defined as the difference between the maximum and minimum frequency across studies). We further excluded variants analyzed in the UK Biobank only or variants not analyzed in the EADB-TOPMed dataset.

Genomic inflation factor lambda was computed with the R package GenABEL⁴⁸ using the median approach after exclusion of the *APOE* region (44 Mb to 46 Mb on chromosome 19 in GRCh38). The linkage disequilibrium (LD) score (LDSC) regression intercept was computed with the LDSC software using the “baselineLD” LD scores built from 1000 Genomes Phase 3⁸. The analysis was restricted to HapMap 3 variants and excluded multi-allelic variants, variants without an rsID and variants in the *APOE* region.

Definition of associated loci. A region of +/- 500kb was defined around each variant with a Stage I P value below 1×10^{-5} . Those regions were then merged with the bedtools software to define non-overlapping regions. The region corresponding to the *APOE* locus was excluded.

We then applied the PLINK clumping procedure to define independent hits in each of those regions. The clumping procedure was applied on all variants with a Stage I P value below 1×10^{-5} . It is an iterative process beginning with the variant with the lowest P value, named index variant. Variants with a Stage I P value below 1×10^{-5} , located within 500 kb of this index variant, and in LD with the index variant (r^2 above 0.001) are assigned to the clump of this index variant. The clumping procedure is then applied on all the remaining variants, until no variant is left. LD was computed in the EADB-TOPMed dataset using high quality (probability ≥ 0.8) imputed genotypes.

Stage II analyses. Variants with a Stage I P value below 1×10^{-5} were sent for follow-up (see Supplementary Information). A fixed-effect meta-analysis was performed with METAL (inverse variance weighted approach) to combine the results across Stage I and Stage II. In each clump, we then reported the replicated variant (same direction of effects between Stage I and Stage II, with a Stage II P value below 0.05) with the lowest P value in the meta-analysis of the Stages I + II. Those variants were considered associated at the genome-wide significance level if they had a P value below 5×10^{-8} in the Stages I + II meta-analysis. Among them, we excluded the variant chr6:32657066:G:A because its frequency amplitude was large.

Pathway analysis. The assignment of Gene Ontology (GO) terms to human genes was obtained from the “gene2go” file, downloaded from NCBI on March 11th 2020. “Parent” GO terms were assigned to genes using the ontology file downloaded from the Gene Ontology website on the same date. GO terms were assigned to genes based on experimental or curated evidence of a specific type, so evidence codes IEA (electronic annotation), NAS (non-traceable author statement), RCA (inferred from reviewed computational analysis) were excluded. Pathways were downloaded from the Reactome website on April 26th 2020. Biocarta, KEGG and Pathway Interaction Database (PID) pathways were downloaded from v7.1 (March 2020) of the Molecular Signatures Database. Analysis was restricted to GO terms containing between 10 and 2000 genes. No size restrictions were placed on the other gene sets, since there were many fewer of them. This resulted in a total of 10,271 gene sets for analysis. Gene set enrichment analyses were performed in MAGMA⁴⁹, correcting for the number of variants in each gene, linkage disequilibrium (LD) between variants and LD between genes. LD was computed in the EADB-TOPMed dataset using high quality (probability ≥ 0.9) imputed genotypes. The measure of pathway enrichment is the MAGMA “competitive” test (where the association statistic for genes in the pathway is compared to those of all other protein-coding genes), as recommended by De Leeuw et al.⁵⁰. We used the “mean” test statistic, which uses the sum of $-\log(\text{variant P value})$ across all genes as the association statistic for genes. The primary analysis assigned variants to genes if they lie within the gene boundaries, but a secondary analysis used a window of 35kb upstream and 10kb downstream to assign variants to genes, as in Kunkle et al.⁶. The primary analysis used all variants with imputation quality above 0.8. We used q-values⁵¹ to account for multiple testing throughout this report.

QTLs/TWAS/MetaMeth. In order to prioritize candidate genes in the new loci, we employed several approaches: (i) expression quantitative trait loci (eQTLs) and colocalization (eQTL coloc) analyses combined with expression transcriptome-wide association studies (eTWAS) in AD-relevant brain regions; (ii) splicing quantitative trait loci (sQTL) and colocalization (sQTL coloc) analyses combined with splicing transcriptome-wide association studies (sTWAS) in AD-relevant brain regions; (iii) genetic-driven methylation as a biological mediator of genetic signals in blood (MetaMeth). In our regions of interest, we systematically searched if a gene has a significant e/sQTL, colocalization e/sTWAS and/or MetaMeth signal(s) within a region of 1 Mb around the lead variant. In addition to the “nearest” genes from the lead variant, we kept for further analyses those exhibiting such AD-related modulations. We then added several additional approaches: (i) data from a genome-wide, high-content siRNA screening approach to assess the functional impact of gene under-

expression on APP metabolism²⁰; (ii) methylation QTL (mQTL) and histone acetylation QTL (haQTL) effects of the lead variants in DLPFC²¹ and (iii) additional eQTL effects of these variants in monocytes and macrophages^{22–27}. A full description of how the genes were prioritized is reported in the Supplementary Information (see also Supplementary Tables 12–21 and Supplementary Fig. 26–34).

Cell type expression. Assignment of newly identified AD risk genes to specific cell classes of the adult brain was performed as previously described⁵². Briefly, middle temporal gyrus (MTG) single-nucleus transcriptomes (15,928 total nuclei derived from 8 human tissue donors ranging in age from 24–66 year), were used to annotate and select 6 main cell classes using *Seurat* 3.1.1⁵³: Glutamatergic Neurons, GABAergic Neurons, Astrocytes, Oligodendrocytes, Microglia and Endothelial cells.

PRS analysis. Twelve longitudinal MCI cohorts were included in the analysis and are fully described in the Supplementary Information and Supplementary Table 23.

PRS were calculated as previously described³⁴. Briefly, we considered 60 variants with genome-wide significant evidence of association with AD in our study (Figure 1, Tables 1 and 2) with a MAF ≥ 0.05 in our case-control study. Variants were directly genotyped or imputed ($R^2 \geq 0.3$). We did not include any *APOE* variants in the PRS. The PRS was calculated as the weighted average of the number of risk increasing alleles for each variant. Weights were based on the respective log(OR) obtained in the Stage II since no samples in this stage were included in the MCI study. The PRS was then multiplied by 60, i.e. the number of included variants. Thus, an increase in HR corresponds to carry one additional average risk allele.

All PCs used were generated per cohort, using the same variants that were used on the case/control study PCA. The number of *APOE*-e4 alleles was obtained based on direct genotyping or, if missing, based on genotypes derived from the TOPMed imputations.

The association of the PRS with risk of progression to dementia in patients with MCI was assessed using Cox proportional hazards-regression analysis. First, the effect was analyzed on progression to all-cause dementia (i.e. regardless of clinical dementia subtype). Next, the analysis was focused on MCI patients converting to AD dementia. To this end, all converters to non-AD dementia were excluded from the analysis sample. Two cohorts (HBA, SAN) were excluded due to missing information on the clinical dementia subtype at this stage. Finally, to assess whether the exclusion of non-AD dementia converters affected our results, the analysis was repeated by coding non-AD converters as censored cases.

Each Cox-regression analysis was first performed unadjusted for covariates and then repeated, adjusted for age, sex and the first four principal components to correct for potential population stratification. Furthermore, analyses were additionally controlled for the number of *APOE*-e4 alleles (assuming an additive effect) to assess the independence of the PRS effect from *APOE*. Moreover, the interaction between the PRS and *APOE*-e4 was tested.

URLs:

Bedtools: <https://bedtools.readthedocs.io/en/latest/>

BCFtools: <http://samtools.github.io/bcftools/bcftools.html>

Samtools: <http://www.htslib.org/doc/samtools.html>

gene2go: <ftp://ftp.ncbi.nlm.nih.gov/gene/DATA/>

Gene Ontology: <http://geneontology.org/docs/download-ontology/>

Reactome: <https://reactome.org/download-data>

KEGG and Pathway Interaction Database (PID) pathways: <https://www.gsea-msigdb.org/gsea/msigdb/index.jsp>

AMP-AD rnaSeqReprocessing Study: <https://www.synapse.org/#!Synapse:syn9702085>

MayoRNAseq WGS VCFs: <https://www.synapse.org/#!Synapse:syn11724002>

ROSMAP WGS VCFs: <https://www.synapse.org/#!Synapse:syn11724057>

MSBB WGS VCFs: <https://www.synapse.org/#!Synapse:syn11723899>

GTEx pipeline: <https://github.com/broadinstitute/gtex-pipeline>

Leafcutter: <https://github.com/davidaknowles/leafcutter>
 RegTools: <https://github.com/griffithlab/regtools>
 Enhanced version of FastQTL: <https://github.com/francois-a/fastqtl>
 Picard: <https://broadinstitute.github.io/picard/>
 eQTLGen: <https://www.eqtlgen.org/>
 eQTL Catalogue database: <https://www.ebi.ac.uk/eqtl/>
 Brain xQTL serve: <http://mostafavilab.stat.ubc.ca/xqtl/>
 GTEx v8 eQTL and sQTL catalogues: <https://www.gtexportal.org/>
 coloc: <https://github.com/chr1swallace/coloc>
 FUSION: https://github.com/gusevlab/fusion_twass
 GTEx v8 expression and splicing prediction models: <http://predictdb.org/>
 MetaXcan: <https://github.com/hakyimlab/MetaXcan>
 FOCUS: <https://github.com/bogdanlab/focus>
 qcat: <https://github.com/nanoporetech/qcat>
 minimap2: <https://github.com/lh3/minimap2>
 NanoStat: <https://github.com/wdecoster/nanostat>
 mosdepth: <https://github.com/brentp/mosdepth>
 ggplot2: <https://ggplot2.tidyverse.org/>
 LocusZoom: <https://github.com/statgen/locuszoom-standalone>
 pyGenomeTracks: <https://github.com/deeptools/pyGenomeTracks>
 VCFs of phased biallelic SNV and INDEL variants of 1KG samples (de novo called on GRCh38):
ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/data_collections/1000_genomes_project/release/20190312_biallelic_SNV_and_INDEL/

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Table 1: Summary of association results in Stage I, Stage II and Stage I + II for known loci with a genome-wide significant signal

Table 2: Summary of association results in Stage I, Stage II and Stage I + II for novel loci with a genome-wide significant signal

Table 3: Results for the most 20 significant pathways from MAGMA pathway analysis based on Stage I results

Figure 1: Manhattan plot of the Stage I results. Variants in new loci are highlighted in red. Loci with a genome-wide significant signal are annotated (known loci in black and new loci in red). Variants with P value below 1×10^{-95} are not shown. The red dotted line represents the genome-wide significance level (P value = 5×10^{-8}), while the black dotted line represents the suggestive significance level (P value = 1×10^{-5}).

Figure 2: Top candidate genes in the 31 new genome-wide-significant loci (left panel) and level of expression in different brain cell types (right panel). In order to prioritize candidate genes in the new loci, we considered the “nearest” protein-coding gene from the lead variant and the genes exhibiting AD-related modulations within a region of 1 Mb around the lead variant. The average expression of each gene expressed by at least 10% of cells (pct.exp > 0.1) was rescaled from 0 to 2, allowing the identification of genes expressed by unique or multiple cell classes. Brown squares show significant hit for the respective column, dark green squares indicate the most probable candidate genes, bright greens squares indicate if several genes were retained in the same locus, and silver square indicate a prioritized gene in an independent known locus. The columns demonstrating lead variant m/haQTL effects in DLPFC and lead variant eQTL effects in monocyte and macrophages are annotated for the type of association (mQTL: methylation QTL, haQTL: histone acetylation QTL, mon: monocyte eQTL, mac: macrophage eQTL), and the superscript indicates that the associated feature is annotated for both genes. Arrow indicates level of expression of the candidate genes (full line for the most probable genes and dotted lines are for genes with less strong evidence).

Figure 3: QTL mapping, QTL colocalization, and TWAS results on SEC61G and TSPAN14 loci. (a) An intergenic, distant, and low-frequency cis-eQTL for *EGFR* locus colocalizes (PP4: 0.98) with EADB GWAS signal where genetic downregulation of *EGFR* is significantly associated with lower AD risk. (b) Elucidation of *TSPAN14* locus: through e/sQTLs, the protective signal is associated with decreased *TSPAN14* expression and increased cryptic splicing (within ADAM10-interacting domain) that are confirmed by long-read single-molecule sequencing on cDNA derived from hippocampus (Hipp.), frontal cortex (FC), lymphoblastoid cell lines (LCL). GWAS and e/sQTL association signals are shown as $-\log_{10}(P)$ and cumulative coverage tracks are in $\log_{10}$ scale. The purple dots show the lead variants in the investigated loci, and LD r^2 values were calculated with respect to these lead variants using the 1000 Genomes non-Finnish European LD reference panel prepared for the TWAS analyses.

Figure 4: Association of PRS with the risk to convert from MCI to AD or dementia in 10 cohorts and meta-analysis. PRS was based on the genetic data of 60 variants (see Methods and Supplementary Table 22).

Table 1

Variant ^a	Chr. ^b	Position ^c	Gene ^d	Known locus	Minor/Major allele	MAF ^e	Stage I			Stage II			Stage I + II				
							App. OR ^f	95% CI ^g	P ^h	OR ⁱ	95% CI ^g	P ^h	App. OR ^f	95% CI ^g	P ^h	I ² (%) ^j	P ^k
rs4233366	1	161 189 357	ADAMTS4	ADAMTS4	T/C	0,263	0.95	0.93-0.97	1.3x10 ⁻⁸	0.94	0.91-0.98	1.2x10 ⁻³	0.95	0.93-0.96	5.9x10 ⁻¹¹	0	9.1x10 ⁻¹
rs679515	1	207 577 223	CR1	CR1	T/C	0,188	1.13	1.11-1.16	5.2x10 ⁻³³	1.14	1.10-1.19	2.6x10 ⁻¹²	1.14	1.11-1.16	1.2x10 ⁻⁴³	21.5	2.2x10 ⁻¹
rs6733839	2	127 135 234	BIN1	BIN1	T/C	0,389	1.18	1.16-1.20	6.5x10 ⁻⁹⁰	1.19	1.16-1.23	6.1x10 ⁻²⁸	1.19	1.17-1.20	5.6x10 ⁻¹¹⁶	32.6	1.1x10 ⁻¹
rs10933431	2	233 117 202	INPP5D	INPP5D	G/C	0,234	0.92	0.90-0.94	1.0x10 ⁻¹⁷	0.94	0.91-0.98	2.2x10 ⁻³	0.92	0.91-0.94	1.7x10 ⁻¹⁹	0	6.8x10 ⁻¹
rs6846529	4	11 023 507	CLNK	CLNK/HS3ST1	C/T	0,283	1.07	1.05-1.09	1.3x10 ⁻¹³	1.07	1.04-1.11	7.1x10 ⁻⁵	1.07	1.05-1.09	4.1x10 ⁻¹⁷	42.4	4.7x10 ⁻²
rs6605556	6	32 615 322	HLA-DQA1	HLA	G/A	0,161	0.91	0.89-0.93	1.0x10 ⁻¹⁷	0.90	0.85-0.95	1.5x10 ⁻⁴	0.91	0.89-0.92	7.7x10 ⁻²¹	0	7.8x10 ⁻¹
rs10947943	6	41 036 354	UNC5CL	TREM2	A/G	0,142	0.95	0.93-0.97	6.2x10 ⁻⁶	0.92	0.88-0.96	1.4x10 ⁻⁴	0.94	0.92-0.96	8.1x10 ⁻⁹	9.8	3.5x10 ⁻¹
rs143332484	6	41 161 469	TREM2	TREM2	T/C	0,013	1.40	1.30-1.50	6.0x10 ⁻¹⁹	1.72	1.45-2.03	3.9x10 ⁻¹⁰	1.44	1.35-1.55	1.7x10 ⁻²⁶	53.7	1.1x10 ⁻²
rs75932628	6	41 161 514	TREM2	TREM2	T/C	0,003	2.42	2.06-2.84	1.4x10 ⁻²⁷	2.30	1.80-2.94	2.3x10 ⁻¹¹	2.39	2.09-2.73	2.5x10 ⁻³⁷	0	6.9x10 ⁻¹
rs60755019	6	41 181 270	TREML2	TREM2	G/A	0,004	1.55	1.32-1.83	1.8x10 ⁻⁷	1.51	1.02-2.25	4.1x10 ⁻²	1.55	1.33-1.80	2.1x10 ⁻⁸	49.9	7.6x10 ⁻²
rs1385742	6	47 627 419	CD2AP	CD2AP	A/T	0,357	1.06	1.05-1.08	4.7x10 ⁻¹³	1.09	1.06-1.13	1.7x10 ⁻⁸	1.07	1.06-1.09	1.7x10 ⁻¹⁹	0	8.4x10 ⁻¹
rs6976216	7	37 854 896	NME8	NME8	A/G	0,309	0.96	0.94-0.97	2.3x10 ⁻⁷	0.94	0.91-0.97	1.4x10 ⁻⁴	0.95	0.94-0.97	2.2x10 ⁻¹⁰	0	8.9x10 ⁻¹
rs7384878	7	100 334 426	SPDY3	ZCWPW1/NYAP1	C/T	0,310	0.93	0.91-0.94	2.1x10 ⁻¹⁸	0.92	0.89-0.95	1.0x10 ⁻⁶	0.92	0.91-0.94	1.3x10 ⁻²³	0	5.8x10 ⁻¹
rs10265814	7	143 424 528	EPHA1	EPHA1	C/G	0,499	1.06	1.04-1.07	8.9x10 ⁻¹²	1.05	1.02-1.08	1.7x10 ⁻³	1.06	1.04-1.07	6.5x10 ⁻¹⁴	49.8	2.1x10 ⁻²
rs73223431	8	27 362 470	PTK2B	PTK2B	T/C	0,369	1.07	1.05-1.09	5.3x10 ⁻¹⁵	1.08	1.04-1.11	3.7x10 ⁻⁶	1.07	1.05-1.09	1.2x10 ⁻¹⁹	0	5.9x10 ⁻¹
rs867230	8	27 610 986	CLU	CLU	C/A	0,397	0.90	0.89-0.92	1.5x10 ⁻³³	0.92	0.89-0.95	1.3x10 ⁻⁷	0.91	0.89-0.92	1.7x10 ⁻³⁹	18.2	2.6x10 ⁻¹
rs7912495	10	11 676 714	USP6NL	ECHDC3	G/A	0,462	1.06	1.04-1.08	2.9x10 ⁻¹²	1.07	1.04-1.11	8.2x10 ⁻⁶	1.06	1.05-1.08	1.5x10 ⁻¹⁶	19.6	2.4x10 ⁻¹
rs10437655	11	47 370 397	SP1	CELF1	A/G	0,399	1.06	1.04-1.08	8.2x10 ⁻¹²	1.05	1.02-1.08	1.5x10 ⁻³	1.06	1.04-1.07	5.3x10 ⁻¹⁴	12.9	3.1x10 ⁻¹
rs1582763	11	60 254 475	MS4A4A	MS4A4A	A/G	0,371	0.92	0.90-0.93	1.7x10 ⁻²⁴	0.88	0.86-0.91	1.2x10 ⁻¹⁴	0.91	0.90-0.92	1.4x10 ⁻³⁶	37.1	8.0x10 ⁻²
rs10792832	11	86 156 833	PICALM	PICALM	A/G	0,358	0.90	0.89-0.91	6.3x10 ⁻³⁶	0.89	0.86-0.92	1.5x10 ⁻¹³	0.90	0.88-0.91	1.1x10 ⁻⁴⁷	19.8	2.4x10 ⁻¹
rs11218343	11	121 564 878	SORL1	SORL1	C/T	0,039	0.85	0.81-0.88	1.0x10 ⁻¹⁴	0.79	0.73-0.86	6.5x10 ⁻⁸	0.84	0.81-0.87	1.0x10 ⁻²⁰	5.2	3.9x10 ⁻¹
rs17125924	14	52 924 962	FERMT2	FERMT2	G/A	0,089	1.09	1.06-1.12	5.8x10 ⁻¹⁰	1.09	1.04-1.15	9.7x10 ⁻⁴	1.09	1.07-1.12	2.2x10 ⁻¹²	0	9.6x10 ⁻¹
rs12590654	14	92 472 511	SLC24A4	SLC24A4/RIN3	A/G	0,328	0.93	0.92-0.95	2.1x10 ⁻¹⁵	0.92	0.89-0.95	1.1x10 ⁻⁶	0.93	0.92-0.94	1.5x10 ⁻²⁰	0	8.2x10 ⁻¹
rs8032980	15	50 711 649	SPPL2A	SPPL2A	C/G	0,345	0.96	0.95-0.98	4.7x10 ⁻⁶	0.94	0.91-0.97	3.6x10 ⁻⁴	0.96	0.94-0.97	1.1x10 ⁻⁸	19.1	2.5x10 ⁻¹
rs593742	15	58 753 575	ADAM10	ADAM10	G/A	0,295	0.94	0.92-0.96	1.0x10 ⁻¹¹	0.94	0.91-0.97	4.0x10 ⁻⁴	0.94	0.93-0.96	1.8x10 ⁻¹⁴	0	6.6x10 ⁻¹
rs117618017	15	63 277 703	APH1B	APH1B	T/C	0,144	1.12	1.09-1.15	1.7x10 ⁻²¹	1.11	1.06-1.17	4.9x10 ⁻⁵	1.12	1.09-1.14	4.3x10 ⁻²⁵	0	6.4x10 ⁻¹
rs889555	16	31 111 250	BCKDK	KAT8	T/C	0,281	0.95	0.93-0.96	1.0x10 ⁻⁹	0.96	0.93-1.00	3.1x10 ⁻²	0.95	0.93-0.96	1.4x10 ⁻¹⁰	6.9	3.8x10 ⁻¹
rs4985556	16	70 660 097	IL34	IL34	A/C	0,115	1.06	1.03-1.09	5.6x10 ⁻⁶	1.13	1.07-1.18	4.1x10 ⁻⁶	1.07	1.05-1.10	9.4x10 ⁻¹⁰	25.9	1.8x10 ⁻¹
rs12446759	16	81 739 398	PLCG2	PLCG2	G/A	0,403	0.94	0.93-0.96	3.6x10 ⁻¹²	0.94	0.91-0.97	3.1x10 ⁻⁴	0.94	0.93-0.96	4.9x10 ⁻¹⁵	17.8	2.6x10 ⁻¹
rs72824905	16	81 908 423	PLCG2	PLCG2	G/C	0,008	0.74	0.67-0.81	2.7x10 ⁻¹⁰	0.80	0.64-0.99	3.7x10 ⁻²	0.75	0.68-0.81	3.6x10 ⁻¹¹	0	5.4x10 ⁻¹
rs7225151	17	5 233 752	SCIMP	SCIMP/RABEP1	A/G	0,124	1.09	1.06-1.12	2.6x10 ⁻¹²	1.08	1.03-1.13	7.0x10 ⁻⁴	1.09	1.07-1.11	7.8x10 ⁻¹⁵	0	5.2x10 ⁻¹
rs199515	17	46 779 275	WNT3	MAPT	G/C	0,219	0.94	0.93-0.96	6.0x10 ⁻⁹	0.93	0.90-0.97	3.7x10 ⁻⁴	0.94	0.93-0.96	1.0x10 ⁻¹¹	62.3	1.5x10 ⁻³
rs616338	17	49 219 935	ABI3	ABI3	T/C	0,012	1.34	1.24-1.45	5.4x10 ⁻¹³	1.27	1.00-1.60	4.8x10 ⁻²	1.33	1.24-1.44	8.3x10 ⁻¹⁴	30.5	1.4x10 ⁻¹
rs2526377	17	58 332 680	TSPOAP1	TSPOAP1	G/A	0,445	0.96	0.94-0.97	4.1x10 ⁻⁸	0.91	0.89-0.94	1.3x10 ⁻⁸	0.95	0.93-0.96	6.6x10 ⁻¹⁴	45.9	3.1x10 ⁻²
rs4277405	17	63 471 557	ACE	ACE	C/T	0,384	0.93	0.92-0.95	7.2x10 ⁻¹⁶	0.95	0.92-0.98	6.5x10 ⁻⁴	0.94	0.92-0.95	2.5x10 ⁻¹⁸	0	8.6x10 ⁻¹
rs12151021	19	1 050 875	ABCA7	ABCA7	A/G	0,336	1.11	1.09-1.13	4.1x10 ⁻³⁰	1.09	1.05-1.13	1.3x10 ⁻⁶	1.11	1.09-1.13	4.2x10 ⁻³⁵	47.7	2.8x10 ⁻²
rs17462136	20	56 412 160	CASS4	CASS4	C/G	0,086	0.89	0.86-0.91	5.8x10 ⁻¹⁶	0.89	0.84-0.94	1.9x10 ⁻⁵	0.89	0.87-0.91	5.5x10 ⁻²⁰	11.0	3.3x10 ⁻¹
rs2830489	21	26 775 872	ADAMTS1	ADAMTS1	T/C	0,281	0.95	0.93-0.96	1.7x10 ⁻⁹	0.94	0.91-0.97	4.5x10 ⁻⁴	0.95	0.93-0.96	3.4x10 ⁻¹²	0	5.6x10 ⁻¹

^aReference SNP (rs) number according to dbSNP build 153; ^bChromosome; ^cGrCh38 assembly; ^dNearest protein coding gene according to Gencode release 33; ^eWeighted average of minor allele frequency across discovery studies; ^fApproximate odds-ratio calculated with respect to the minor allele; ^g95% confidence interval; ^hP value; ⁱOdds-ratio calculated with respect to the minor allele; ^jHeterogeneity statistic; ^kP value for heterogeneity statistic

Table 2

Locus number	Variant ^a	Chr. ^b	Position ^c	Gene ^d	Major/Minor allele	MAF ^e	Stage I			Stage II			Stage I + II				
							App. OR ^f	95% CI ^g	P ^h	OR ⁱ	95% CI ^g	P ^h	App. OR ^f	95% CI ^g	P ^h	I ² (%) ^j	P ^k
1	rs143080277	2	105 749 599	NCK2	C/T	0.005	1.48	1.31-1.66	7.9x10 ⁻¹¹	1.42	1.13-1.78	2.3x10 ⁻³	1.47	1.32-1.63	7.1x10 ⁻¹³	20	2.7x10 ⁻¹
2	rs61762319	3	155 084 189	MME	G/A	0.026	1.15	1.10-1.21	2.1x10 ⁻⁸	1.30	1.15-1.47	2.6x10 ⁻⁵	1.17	1.12-1.23	1.3x10 ⁻¹¹	5.5	3.9x10 ⁻¹
3	rs3822030	4	993 555	IDUA	G/T	0.429	0.95	0.93-0.97	5.0x10 ⁻¹⁰	0.96	0.93-0.99	4.3x10 ⁻³	0.95	0.94-0.96	8.3x10 ⁻¹²	2.7	4.2x10 ⁻¹
4	rs250435	5	14 664 712	OTULIN	G/A	0.079	1.08	1.05-1.12	4.2x10 ⁻⁷	1.13	1.07-1.21	6.2x10 ⁻⁵	1.09	1.06-1.13	2.5x10 ⁻¹⁰	35.9	8.8x10 ⁻¹
5	rs62375397	5	87 002 714	RASA1	T/C	0.210	1.08	1.06-1.10	8.6x10 ⁻¹⁴	1.06	1.02-1.10	1.3x10 ⁻³	1.07	1.06-1.09	5.2x10 ⁻¹⁶	45.3	3.3x10 ⁻¹
6	rs113706587	5	180 201 150	RASGEF1C	A/G	0.110	1.10	1.07-1.13	3.4x10 ⁻¹²	1.10	1.05-1.16	1.9x10 ⁻⁴	1.10	1.07-1.12	2.8x10 ⁻¹⁵	0	8.4x10 ⁻¹
7	rs785129	6	114 291 731	HS3ST5	T/C	0.350	1.05	1.03-1.06	2.9x10 ⁻⁷	1.04	1.00-1.07	3.1x10 ⁻²	1.04	1.03-1.06	3.0x10 ⁻⁸	10.6	3.4x10 ⁻¹
8	rs6943429	7	7 817 263	UMAD1	T/C	0.421	1.04	1.03-1.06	3.0x10 ⁻⁷	1.06	1.02-1.09	6.2x10 ⁻⁴	1.05	1.03-1.06	9.0x10 ⁻¹⁰	50.3	1.6x10 ⁻¹
9	rs10952097	7	8 204 382	ICA1	T/C	0.114	1.07	1.04-1.10	3.3x10 ⁻⁷	1.08	1.03-1.14	3.1x10 ⁻³	1.08	1.05-1.10	3.8x10 ⁻⁹	52.2	1.4x10 ⁻¹
10	rs7781670	7	12 211 934	TMEM106B	G/C	0.409	0.96	0.94-0.98	6.1x10 ⁻⁷	0.96	0.93-0.99	8.6x10 ⁻³	0.96	0.95-0.97	1.7x10 ⁻⁸	42.8	4.5x10 ⁻¹
11	rs76928645	7	54 873 635	SEC61G	T/C	0.103	0.93	0.90-0.95	1.5x10 ⁻⁸	0.95	0.90-1.00	4.9x10 ⁻²	0.93	0.91-0.95	2.9x10 ⁻⁹	21.6	2.2x10 ⁻¹
12	rs4731	8	11 808 828	FDFT1	G/A	0.081	1.08	1.05-1.11	7.6x10 ⁻⁸	1.06	1.00-1.12	4.3x10 ⁻²	1.08	1.05-1.11	1.2x10 ⁻⁸	54.4	9.7x10 ⁻¹
13	rs13276936	8	94 983 473	NDUFAF6	T/C	0.466	1.05	1.03-1.07	2.8x10 ⁻⁹	1.04	1.01-1.07	1.0x10 ⁻²	1.05	1.03-1.06	1.1x10 ⁻¹⁰	0	5.1x10 ⁻¹
14	rs34173062	8	144 103 704	SHARPIN	A/G	0.081	1.12	1.09-1.16	2.9x10 ⁻¹²	1.17	1.08-1.27	1.8x10 ⁻⁴	1.13	1.09-1.16	3.7x10 ⁻¹⁵	44.6	4.2x10 ⁻¹
15	rs1800978	9	104 903 697	ABCA1	G/C	0.130	1.07	1.05-1.10	1.6x10 ⁻⁸	1.05	1.00-1.10	3.7x10 ⁻²	1.07	1.04-1.09	2.5x10 ⁻⁹	27.7	1.6x10 ⁻¹
16	rs7068231	10	60 025 170	ANK3	T/G	0.403	0.95	0.94-0.97	6.8x10 ⁻⁹	0.96	0.93-0.99	1.0x10 ⁻²	0.95	0.94-0.97	2.5x10 ⁻¹⁰	0	5.9x10 ⁻¹
17	rs6586028	10	80 494 228	TSPAN14	C/T	0.196	0.92	0.91-0.94	1.3x10 ⁻¹⁴	0.94	0.91-0.98	2.3x10 ⁻³	0.93	0.91-0.94	1.7x10 ⁻¹⁶	0	4.8x10 ⁻¹
18	rs6584063	10	96 266 650	BLNK	G/A	0.046	0.89	0.86-0.93	1.8x10 ⁻⁷	0.85	0.78-0.93	2.6x10 ⁻⁴	0.89	0.85-0.92	3.4x10 ⁻¹⁰	0	9.3x10 ⁻¹
19	rs3919533	12	7 055 497	C1S	C/T	0.172	0.95	0.93-0.97	1.0x10 ⁻⁶	0.94	0.91-0.98	5.7x10 ⁻³	0.95	0.93-0.97	1.9x10 ⁻⁸	28.5	1.6x10 ⁻¹
20	rs74091721	14	106 638 262	IGH gene	T/G	0.145	0.94	0.92-0.96	6.7x10 ⁻⁷	0.94	0.89-0.98	3.1x10 ⁻³	0.94	0.92-0.96	7.7x10 ⁻⁹	0	9.9x10 ⁻¹
21	rs2009833	15	49 901 356	ATP8B4	A/G	0.368	0.95	0.94-0.97	3.1x10 ⁻⁸	0.96	0.93-0.99	5.9x10 ⁻³	0.95	0.94-0.97	6.5x10 ⁻¹⁰	0	6.5x10 ⁻¹
22	rs3848143	15	64 131 307	SNX1	G/A	0.220	1.05	1.03-1.07	1.1x10 ⁻⁶	1.07	1.03-1.11	3.0x10 ⁻⁴	1.05	1.04-1.07	2.0x10 ⁻⁹	0	5.5x10 ⁻¹
23	rs12592898	15	78 936 857	CTSH	A/G	0.133	0.94	0.92-0.97	1.3x10 ⁻⁶	0.93	0.89-0.98	4.5x10 ⁻³	0.94	0.92-0.96	2.1x10 ⁻⁸	3.7	4.1x10 ⁻¹
24	rs9783783	16	30 067 171	ALDOA	C/G	0.492	0.96	0.94-0.97	4.5x10 ⁻⁸	0.97	0.94-1.00	4.8x10 ⁻²	0.96	0.95-0.97	8.1x10 ⁻⁹	0	8.9x10 ⁻¹
25	rs450674	16	79 574 511	MAF	C/T	0.373	0.96	0.94-0.97	1.1x10 ⁻⁷	0.97	0.94-1.00	3.5x10 ⁻²	0.96	0.94-0.97	1.3x10 ⁻⁸	14.1	3.0x10 ⁻¹
26	rs56407236	16	90 103 687	PRDM7	A/G	0.069	1.12	1.08-1.15	1.3x10 ⁻¹¹	1.10	1.02-1.17	9.0x10 ⁻³	1.11	1.08-1.14	4.4x10 ⁻¹³	0	6.0x10 ⁻¹
27	rs5848	17	44 352 876	GRN	T/C	0.289	1.07	1.05-1.09	1.8x10 ⁻¹²	1.04	1.01-1.08	1.2x10 ⁻²	1.06	1.05-1.08	1.2x10 ⁻¹³	0	9.3x10 ⁻¹
28	rs8103552	19	49 955 674	SIGLEC11	A/C	0.240	1.05	1.03-1.07	5.7x10 ⁻⁶	1.09	1.05-1.13	1.2x10 ⁻⁶	1.06	1.04-1.08	2.5x10 ⁻¹⁰	0	4.8x10 ⁻¹
29	rs12984029	19	54 265 512	LILRB5	A/G	0.235	1.06	1.04-1.09	3.2x10 ⁻⁹	1.05	1.01-1.09	9.8x10 ⁻³	1.06	1.04-1.08	1.2x10 ⁻¹⁰	33	1.2x10 ⁻¹
30	rs6742	20	63 743 088	SLC2A4RG	T/C	0.221	0.95	0.93-0.97	2.5x10 ⁻⁶	0.93	0.89-0.98	2.1x10 ⁻³	0.95	0.93-0.97	2.6x10 ⁻⁸	0	9.6x10 ⁻¹
31	rs2154481	21	26 101 558	APP	C/T	0.476	0.95	0.94-0.97	1.0x10 ⁻⁹	0.96	0.93-0.99	7.6x10 ⁻³	0.95	0.94-0.97	3.0x10 ⁻¹¹	5.1	4.0x10 ⁻¹

^aReference SNP (rs) number according to dbSNP build 153; ^bChromosome; ^cGRCh38 assembly; ^dNearest protein coding gene according to Gencode release 33; ^eWeighted average of minor allele frequency across discovery studies; ^fApproximate odds-ratio calculated with respect to the minor allele; ^g95% confidence interval; ^hP value; ⁱOdds-ratio calculated with respect to the minor allele; ^jHeterogeneity statistic; ^kP value for heterogeneity statistic

Table 3

Pathway	Number of genes in the pathway in the dataset	P value	q value ^a	Pathway description
GO:1902992	14	9.5x10 ⁻¹⁷	7.5x10 ⁻¹³	Negative regulation of amyloid precursor protein catabolic process
GO:1902430	12	1.8x10 ⁻¹²	7.3x10 ⁻⁹	Negative regulation of amyloid-beta formation
GO:0048156	12	3.6x10 ⁻¹²	9.4x10 ⁻⁹	Tau protein binding
R-HSA-5619055	1	8.3x10 ⁻¹¹	1.6x10 ⁻⁷	Defective SLC24A4 causes hypomineralized amelogenesis imperfecta (AI)
GO:1905245	11	9.8x10 ⁻¹⁰	1.6x10 ⁻⁶	Regulation of aspartic-type peptidase activity
GO:1902991	32	1.6x10 ⁻⁹	2.1x10 ⁻⁶	Regulation of amyloid precursor protein catabolic process
GO:1902003	27	5.0x10 ⁻⁹	5.7x10 ⁻⁶	Regulation of amyloid-beta formation
GO:0032994	26	1.0x10 ⁻⁸	1.0x10 ⁻⁵	Protein-lipid complex
GO:0033700	12	3.7x10 ⁻⁷	2.9x10 ⁻⁴	Phospholipid efflux
GO:0034358	23	4.1x10 ⁻⁷	2.9x10 ⁻⁴	Plasma lipoprotein particle
GO:1990777	23	4.1x10 ⁻⁷	2.9x10 ⁻⁴	Lipoprotein particle
R-HSA-198933	121	8.2x10 ⁻⁷	5.4x10 ⁻⁴	Immunoregulatory interactions between a Lymphoid and a non-Lymphoid cell
GO:0034364	18	1.2x10 ⁻⁶	7.1x10 ⁻⁴	High-density lipoprotein particle
GO:0042116	28	2.0x10 ⁻⁶	1.2x10 ⁻³	Macrophage activation
GO:0071825	41	2.8x10 ⁻⁶	1.5x10 ⁻³	Protein-lipid complex subunit organization
GO:0001774	18	3.2x10 ⁻⁶	1.5x10 ⁻³	Microglial cell activation
GO:0002269	18	3.2x10 ⁻⁶	1.5x10 ⁻³	Leukocyte activation involved in inflammatory response
GO:0033003	24	3.8x10 ⁻⁶	1.7x10 ⁻³	Regulation of mast cell activation
GO:0045332	22	4.3x10 ⁻⁶	1.7x10 ⁻³	Phospholipid translocation
GO:0044433	1132	4.4x10 ⁻⁶	1.7x10 ⁻³	Cytoplasmic vesicle part

^asignificant after FDR correction (q value ≤ 0.05)

Figure 1

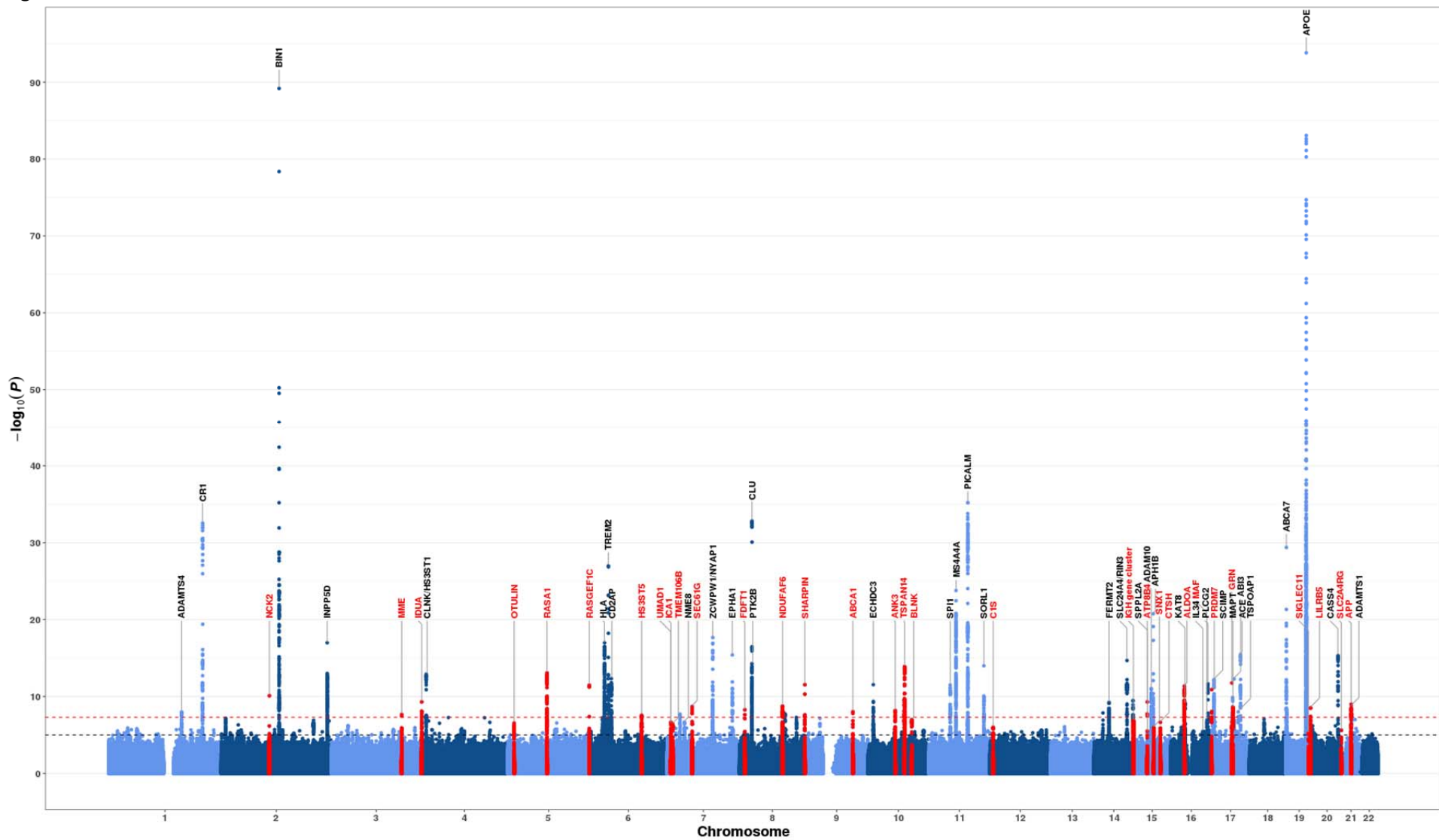


Figure 2

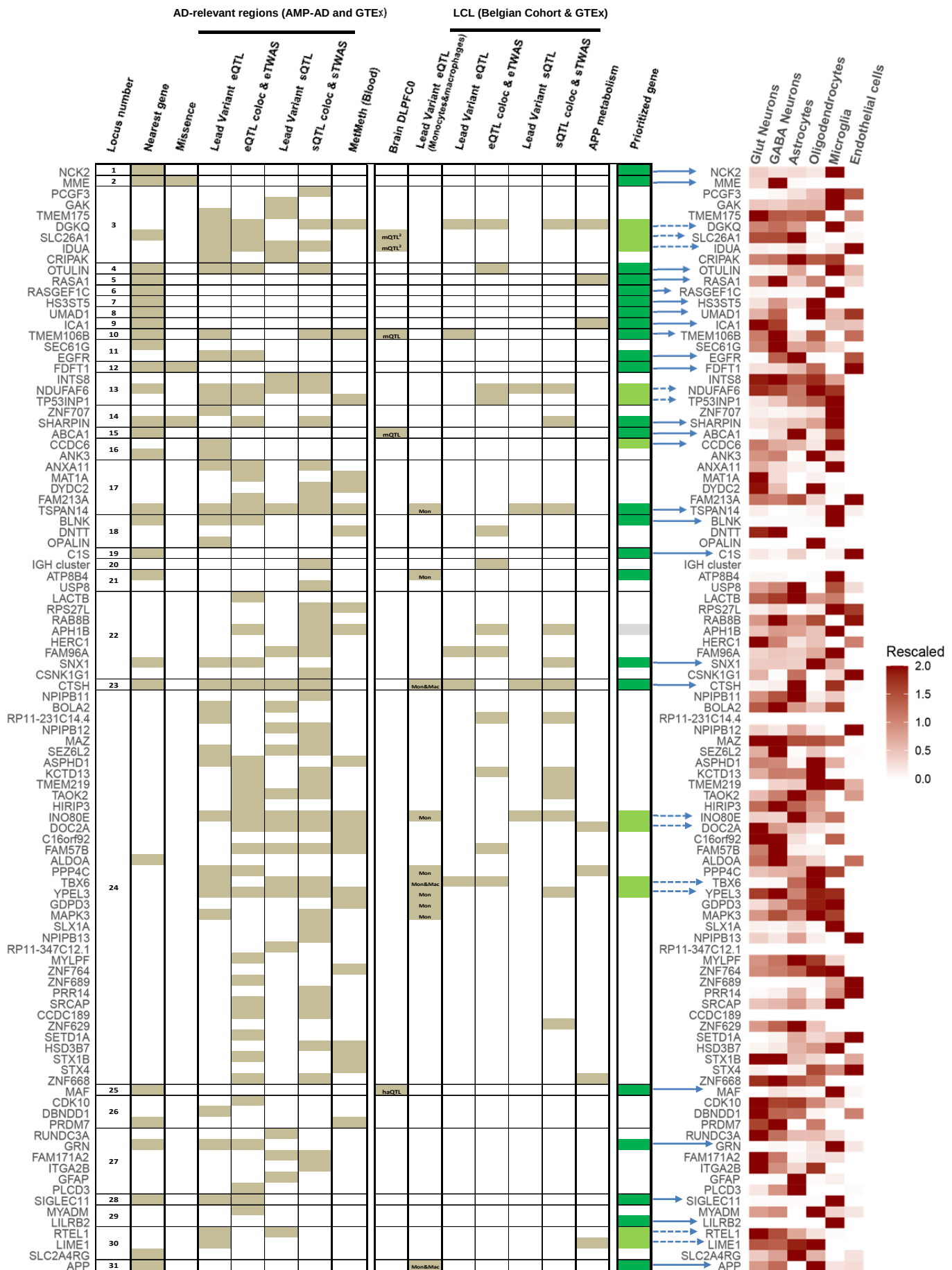


Figure 3 :

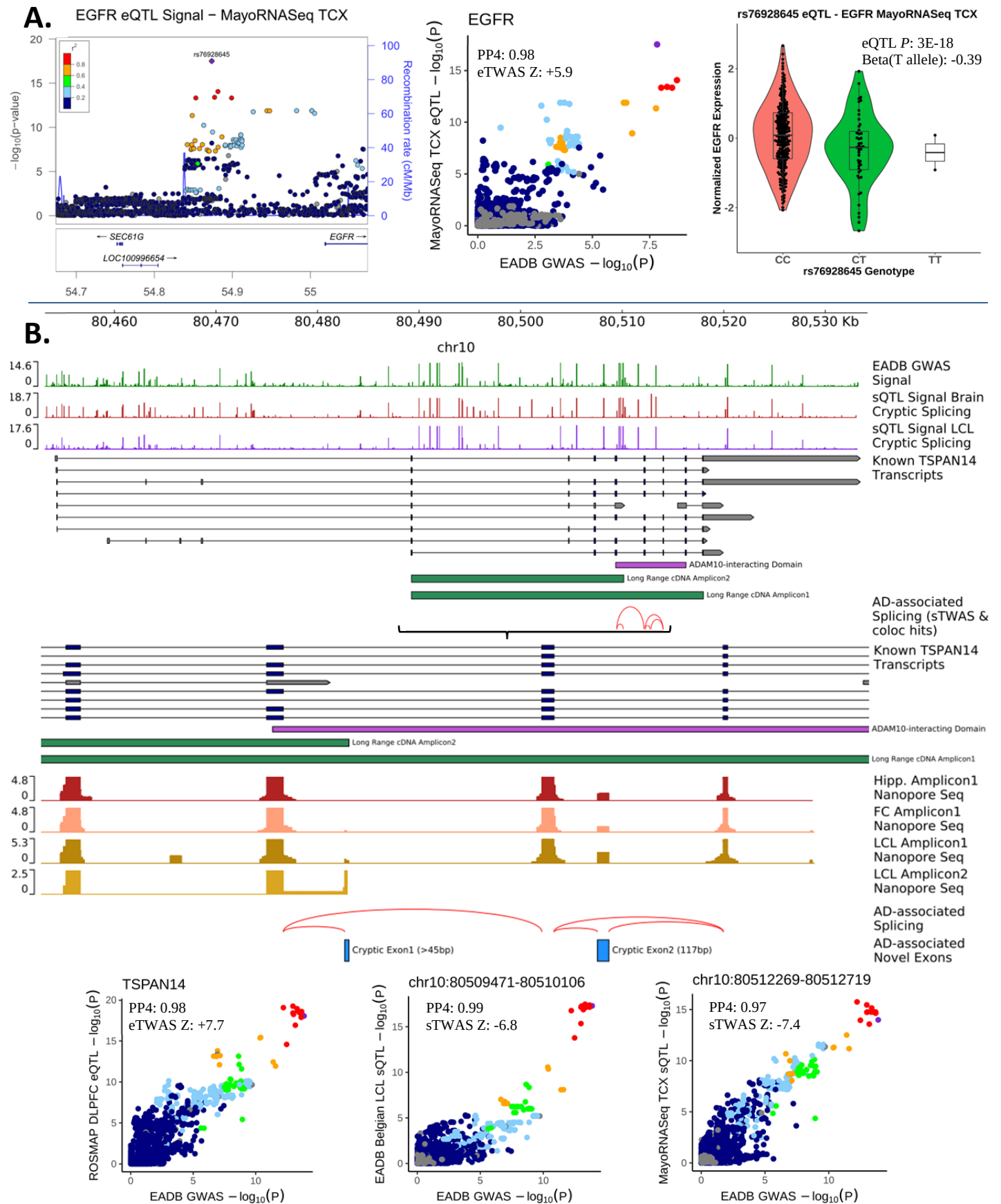


Figure 4

